

Reproduction of single-cell RNA-seq workflow

Associated website: <https://scanpy.readthedocs.io/en/latest/tutorials/basics/clustering-2017.html>

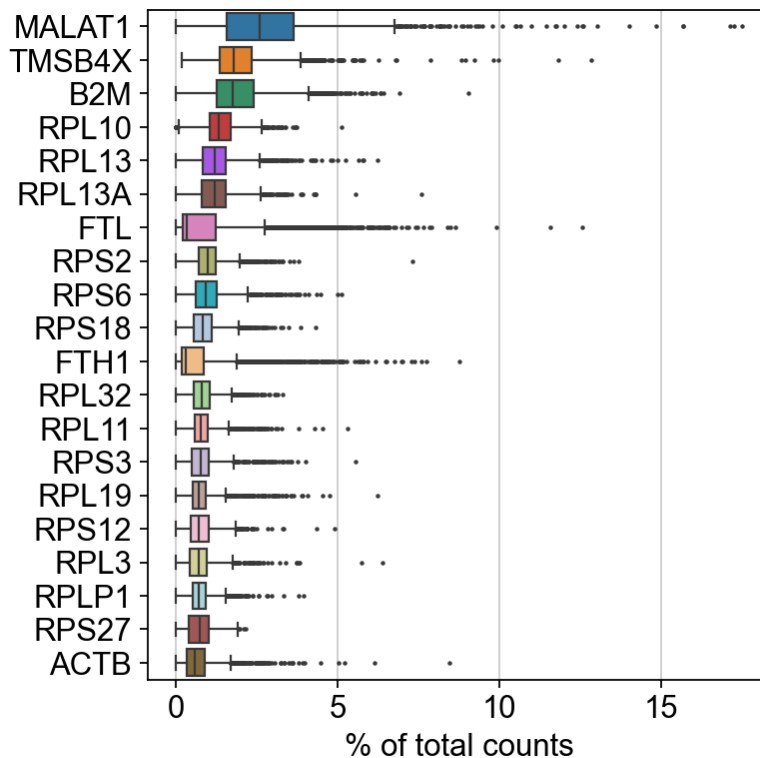
Data source: 3k PBMCs from a Healthy Donor and are freely available from 10x Genomics

1. Processing

Our goal for processing is get a general idea of our data structure.

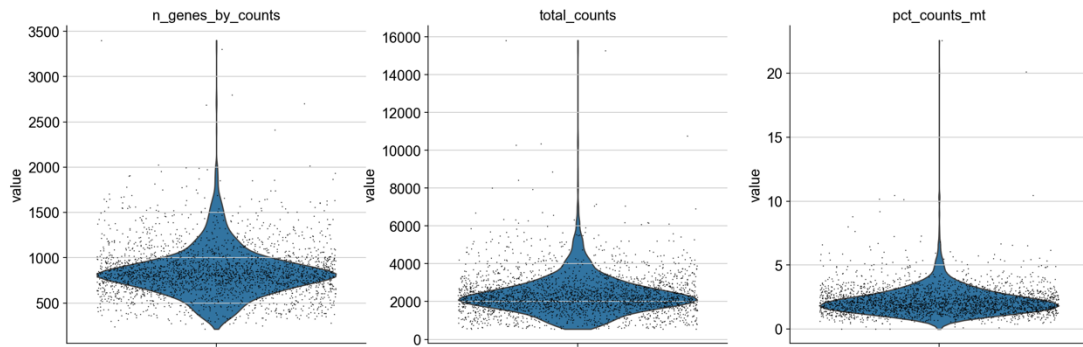
Genes with the highest fraction of total gene counts in a single cell.

For example $\%MALAT1 = \text{counts of MALAT1 in a single cell} / \text{counts of all genes in a single cell}$



The three graphs below correspond to:

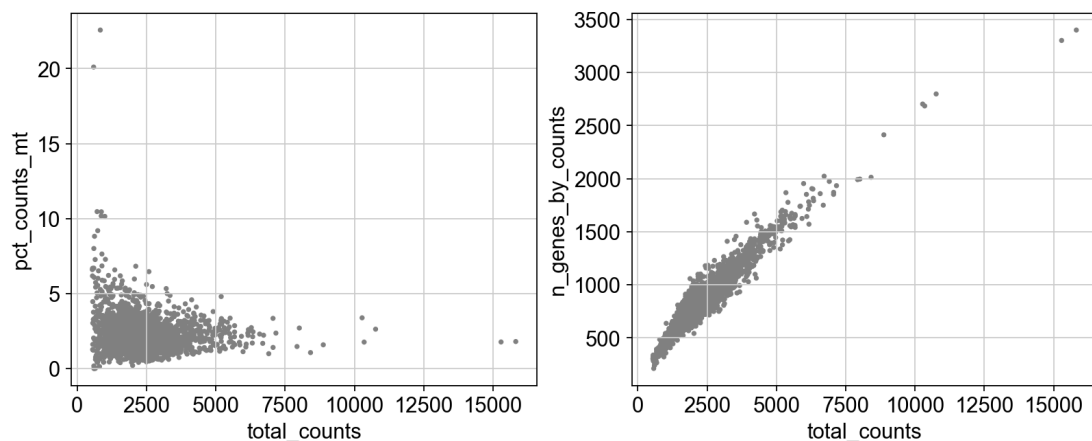
- The total count of a specific gene in all cells
- total counts of all kinds of genes in one single cell
- the fraction of mitochondria RNAs among the total number of genes



The two graphs below shows “pct_counts_mt vs. total counts” and “n_genes_by_counts vs. total_counts”

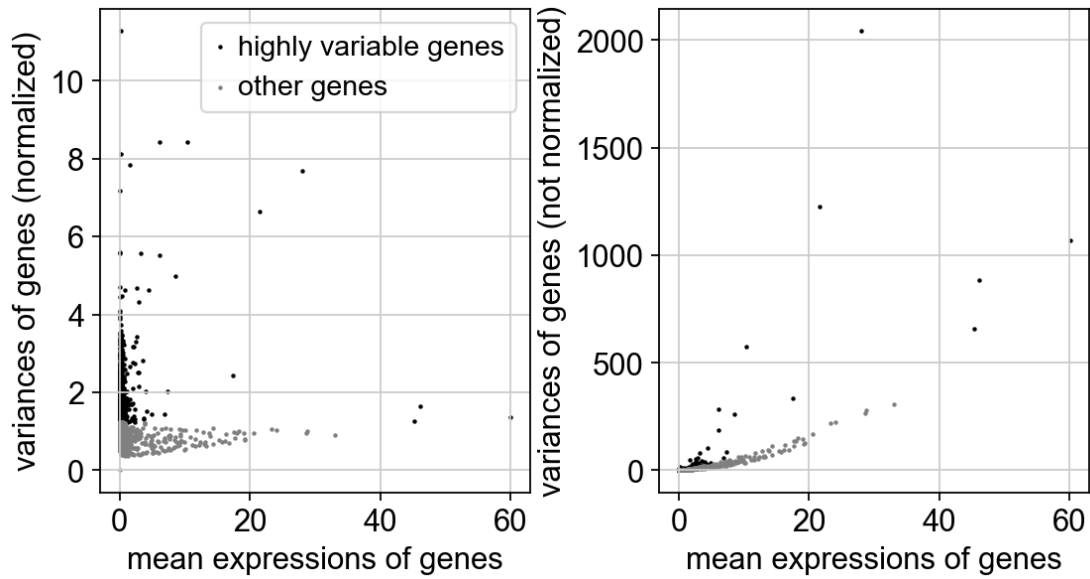
When mitochondria RNAs takes a huge proportion of all counts and total counts number is small, it is very likely that cell is damaged or dying.

The total count and the n_genes_by_counts generally follow a linear relationship. However, there are some outliers that exist on the top-right corner. It is mostly likely that these points refer to doublets or multiplets. Datapoints at the down-left corner refers to dying cells that don't include much RNAs.



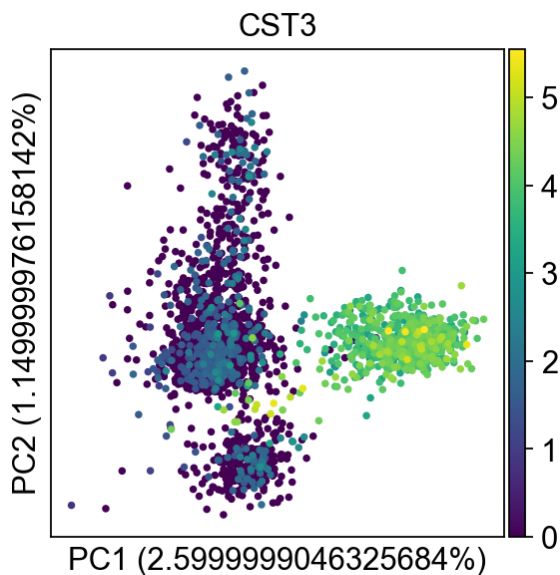
Recognizing the potential outliers in our data, we drop these abnormal values, $(adata.obs.n_genes_by_counts < 2500) \& (adata.obs.n_genes_by_counts > 200) \& (adata.obs.pct_counts_mt < 5)$, normalize the data matrix to 10000 reads per cell and finally, logarithmize the data.

In order to identify high-variable genes, which could be used as an indicator of a special cell, we draw the following plots. The right plot shows the drawback of not normalizing the data, variances among cells with different totals counts can vary greatly.

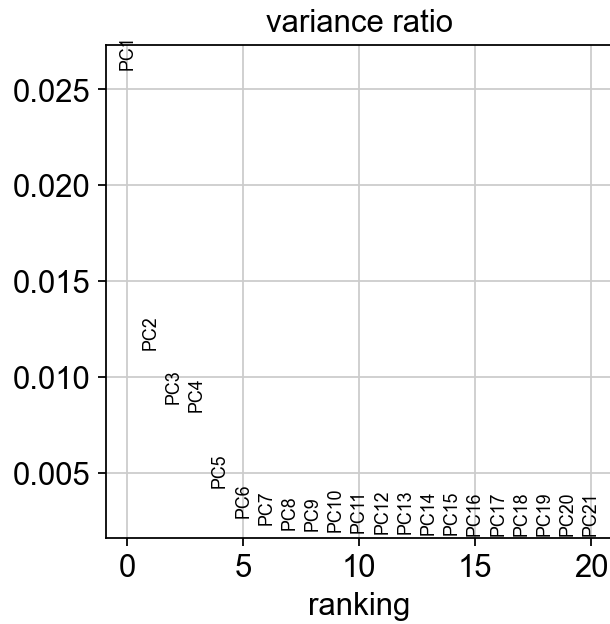


2. Principal component analysis

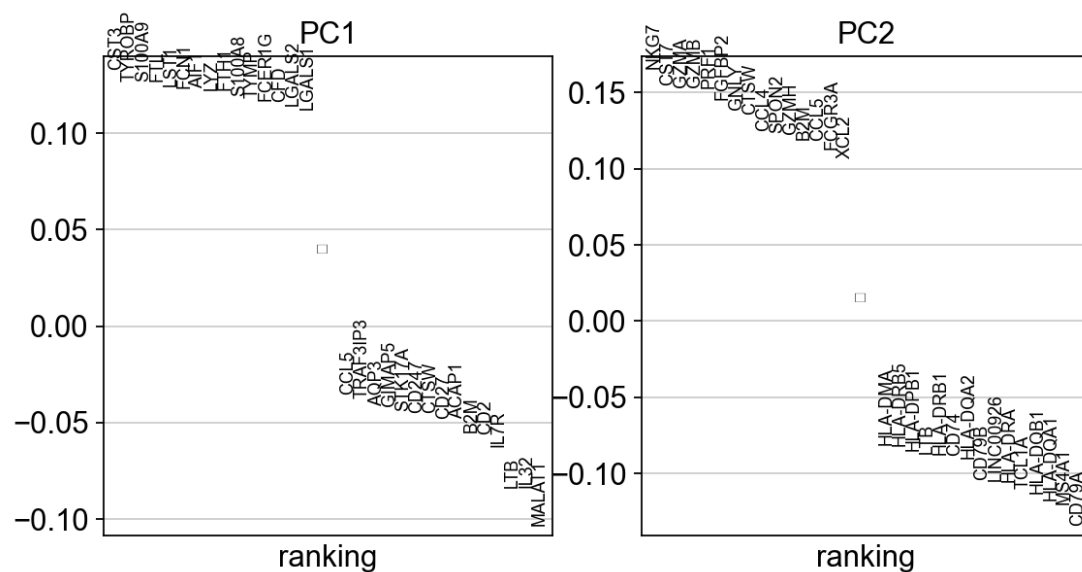
CST3 is a gene randomly selected. Yellow indicates the gene is highly expressed, while blue means the gene is in low expression. Since our data matrix is high-dimensional, it can be especially hard for us to take all those genes as a dimension and plot our data in the space. In the real computing procedure, we created new parameter such as PC1 and PC2. Each parameter is a linear combination of dozens of genes. For example, $PC1 = \theta_0 + \theta_1 gene1 + \theta_2 gene2 + \theta_3 gene3 + \dots$. On the label, the number behind the PC1 indicates how much variability can be explained by PC1.



The following plot is the rank for the most explainable PCs. This gives us information about how many PCs we should consider in order to compute the neighborhood relations of cells.

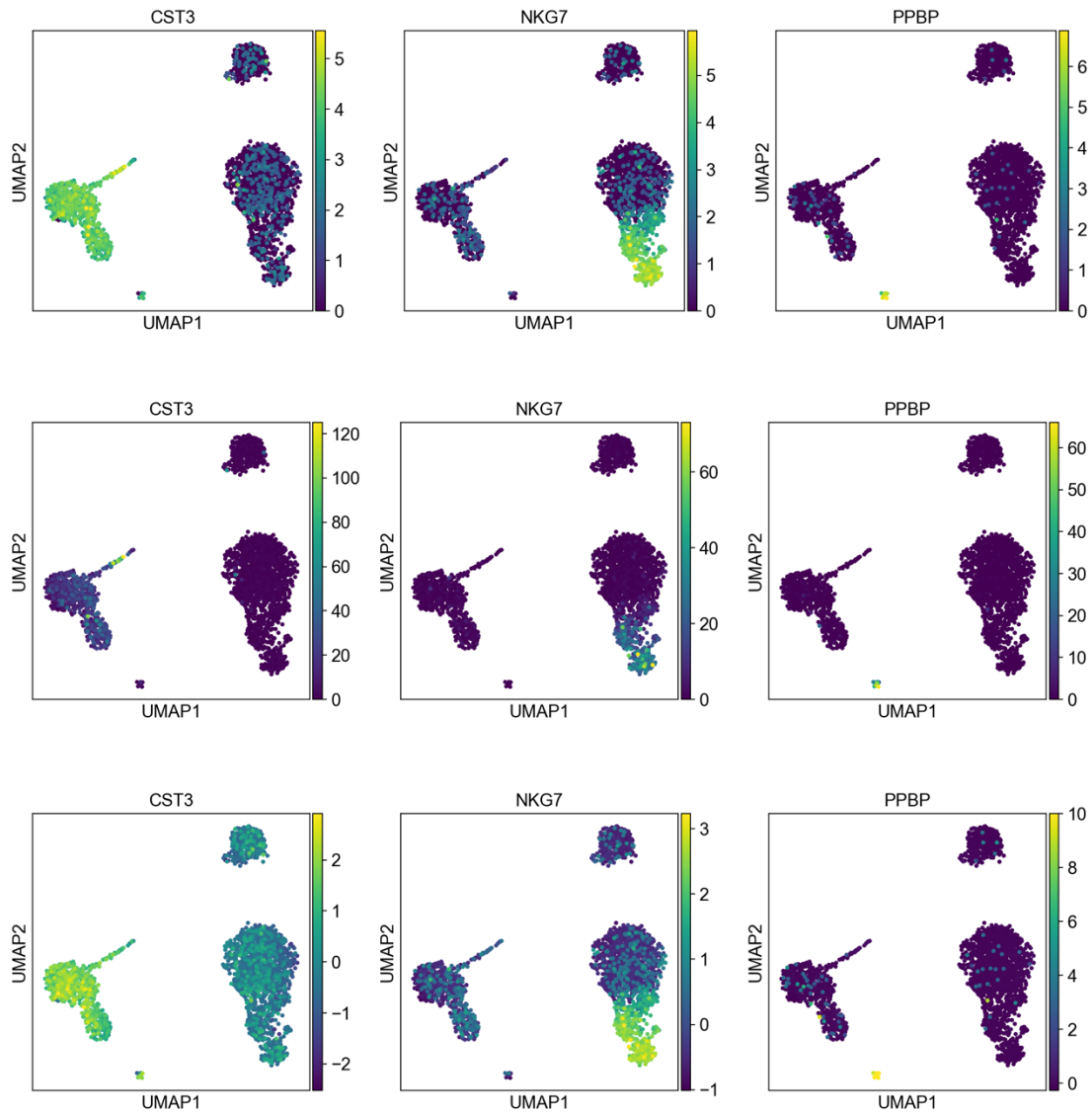


The following displays the most related genes to each PC.



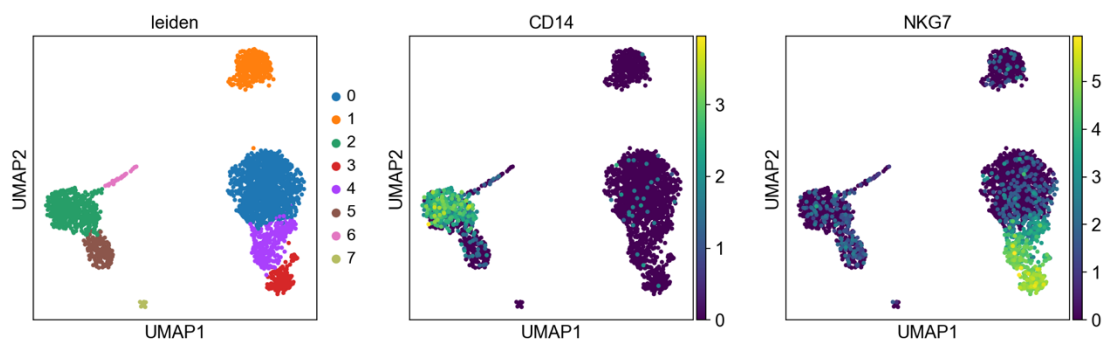
3. Computing and embedding the neighborhood graph

By embedding each cell's PC information into two dimensions using the UMAP function, we can easily identify some cell clusters. Aiming to get a clearer picture of the specific cell function, we color our data with regard to the expression level of a specific gene.



4. Clustering the neighborhood graph

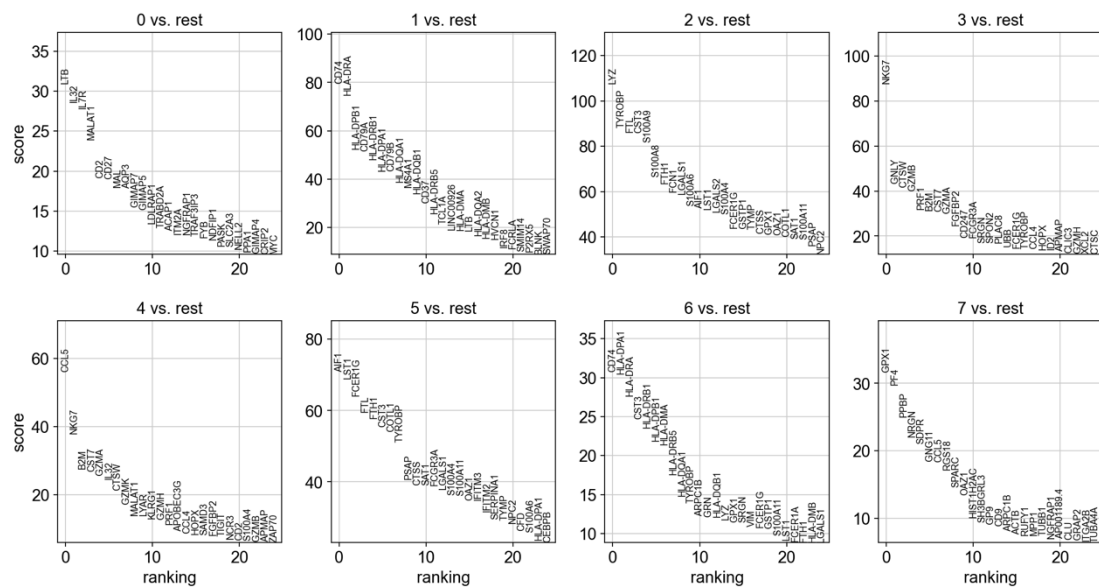
Picture in 2-dimentional space has its drawback for not being able to show cluster relationships. Leiden graph-clustering method (community detection based on optimizing modularity) is recommended to show a more explicit picture of clustering the neighborhood graph of cells, as is shown in Picture 1.



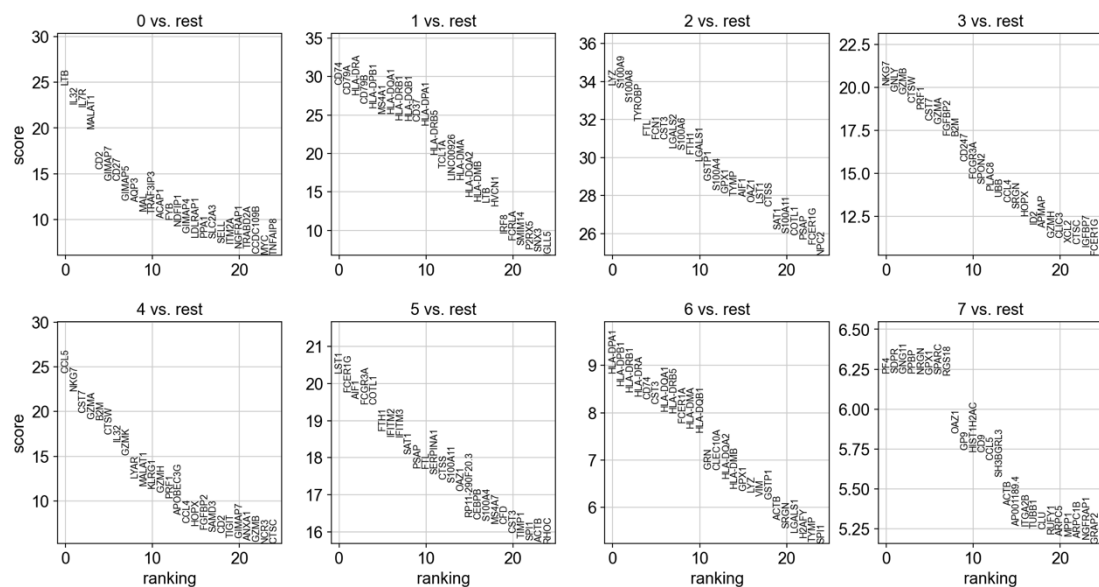
5. Finding marker genes

There are many algorithms to compute the rankings for the highly differential genes in each cluster. The three sets of graphs show maker gene finding by t-test, Wilcoxon, logreg method.

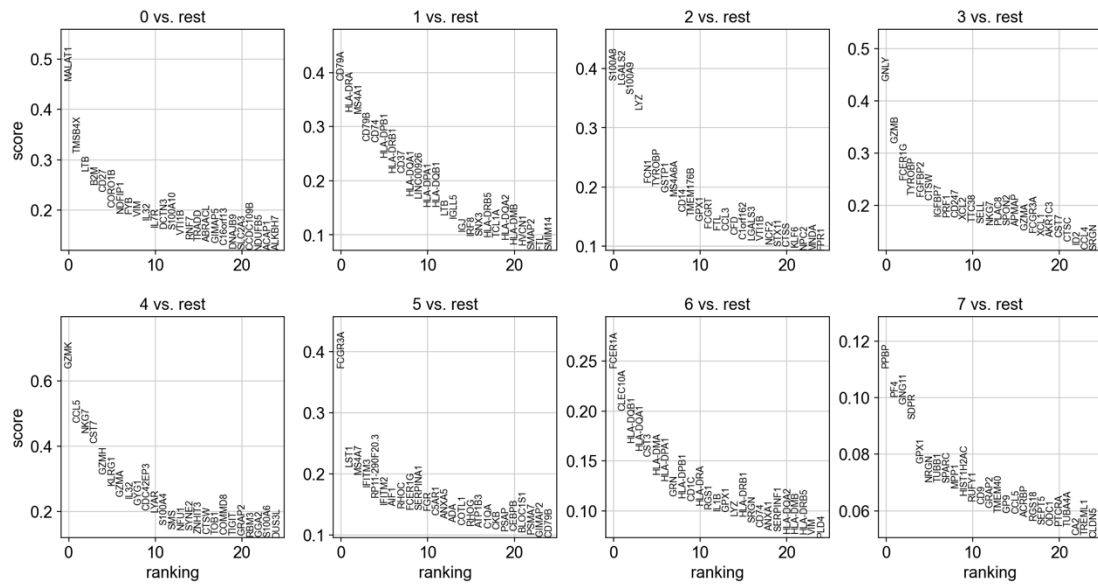
T-test:



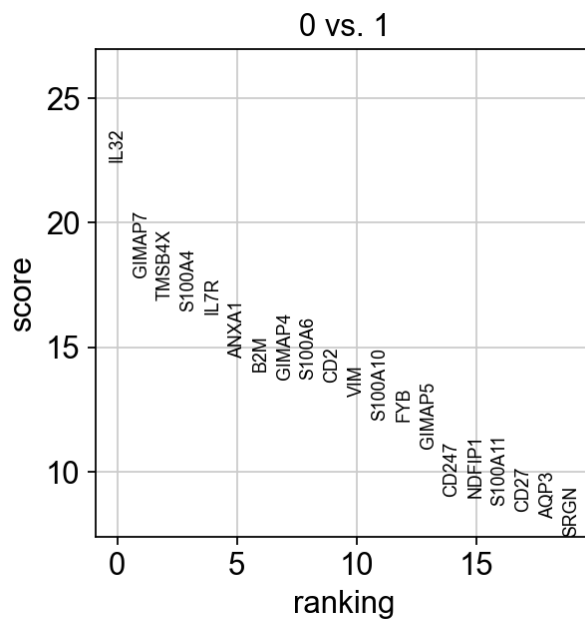
Wilcoxon:



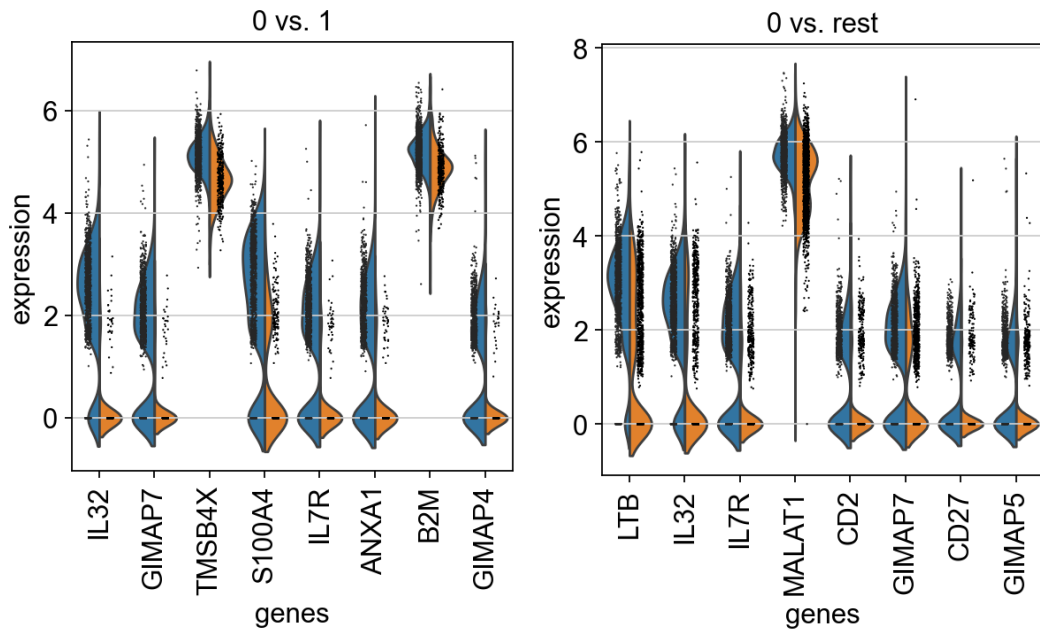
Logreg:



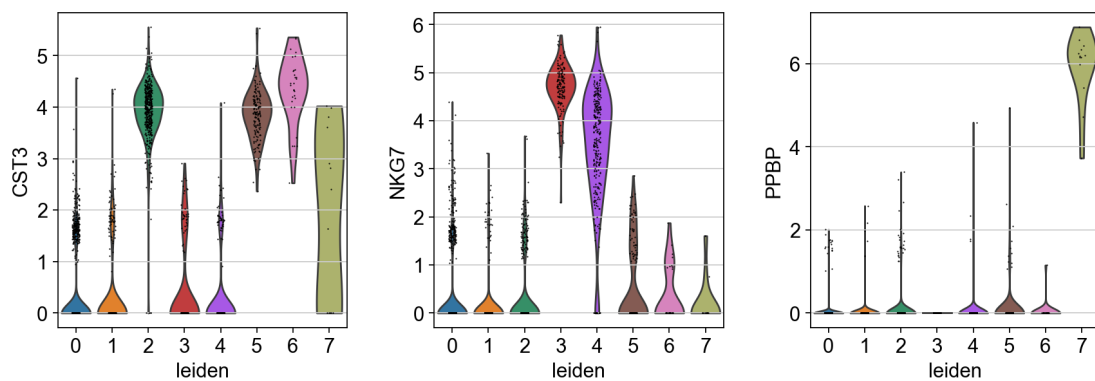
The difference between two single clusters:



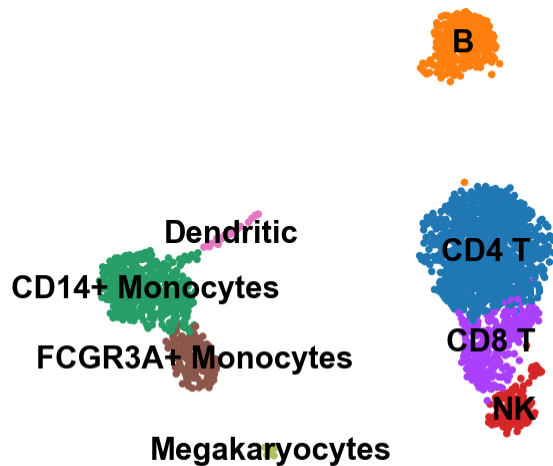
If you want to visualize the detailed difference, you shall use the violin plot. As is shown below. You can not only see the difference of magnitude but also difference in the shape of distribution for each gene's expression.



If you want to compare a certain gene across groups, use the following.



You can infer each clusters' name from its highly expressed genes and name each cluster.



At last, maker genes can be visualized via the following graphs.

